

Computer System

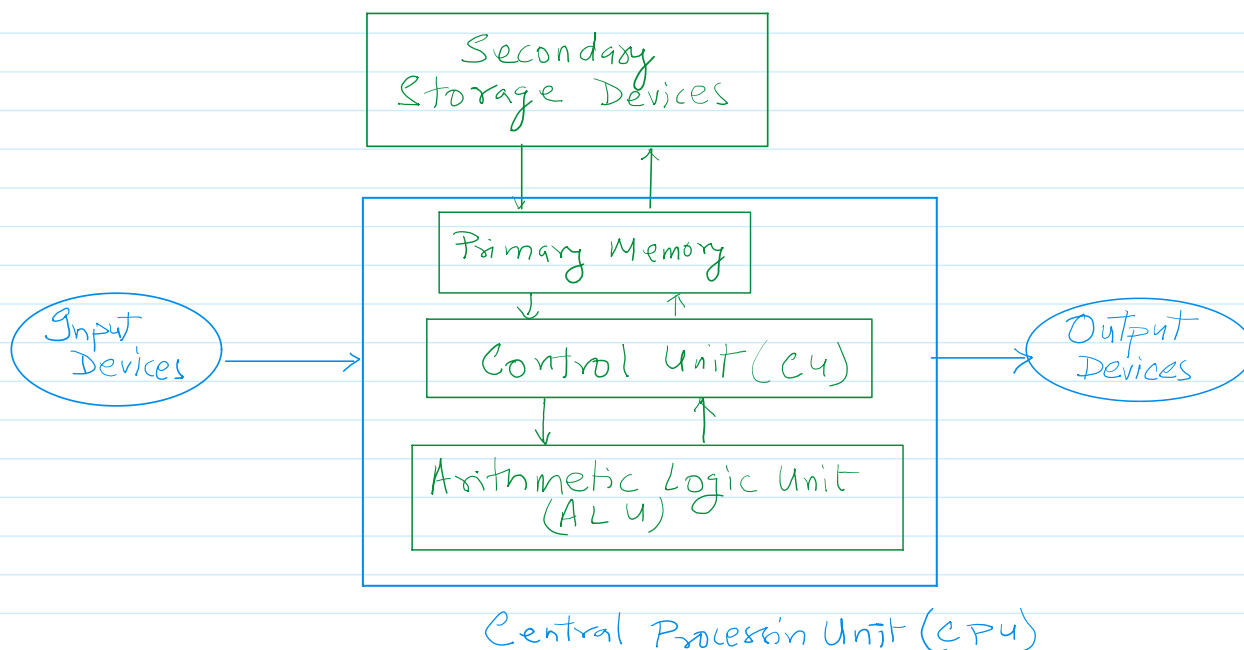
26 July 2026 09:38

A computer is an electronic device that can be programmed to accept data (input) process it, and generate a result (output).

What is computer system?

A computer along with additional hardware and software together is called a **computer system**.

A computer system primarily comprises a central processing unit (CPU) memory, Input/output devices and storage devices. All these components function together as a single unit to deliver the desired output. A computer system comes in various forms and sizes. It can vary from high end server to personal desktop.



Central Processing Unit(CPU)

It is the electronic circuitry of a computer that carries out the actual processing and usually referred as the brain of the computer. It is commonly called processor also. Physically, a CPU can be placed on one or more microchips called integrated circuits (IC). The ICs comprise semiconductor materials.

The CPU is given instruction and data through programs. The CPU then fetches the program and data from the memory and performs arithmetic and logic operations as per the given instructions and store the result back to memory.

While processing the CPU stores the data as well as instructions in its local memory called registers. Registers are part of the CPU chips and they are limited in size and number. Different registers are used for storing data instructions or intermediate results.

Other than registers, the CPU has two main components - Arithmetic logic unit (ALU) and control unit (CU). ALU performs all the arithmetic and logic operations that need to be done as per the instruction in a program. Control unit controls sequential instruction execution, interprets instructions, and guides data flow through the computer's memory ALU and input or output devices. CPU is also popularly known as microprocessors.



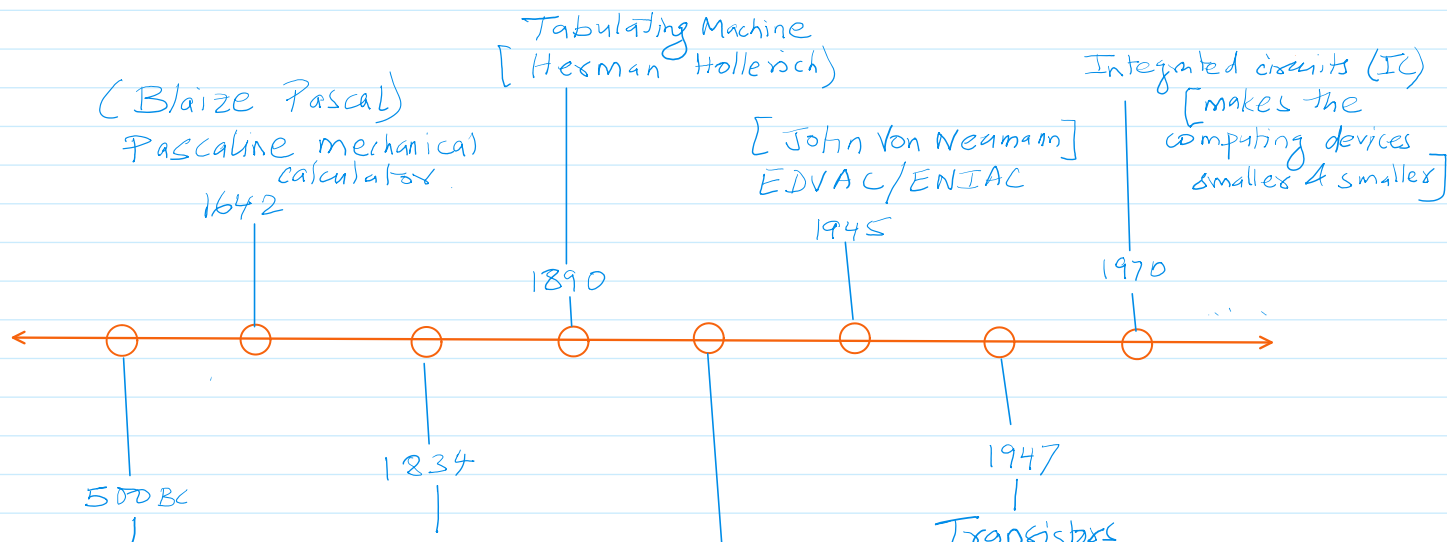
Input Devices: The devices through which control signals are sent to the computer are termed as input devices. These devices convert the input data into a digital form that is acceptable by the computer system. Some examples of input devices include keyboard, mouse, scanner, touch screen, etc.

Data entered through input devices is temporarily stored in main memory (also called RAM) of the computer system. For permanent storage and future use, the data as well as instructions are stored permanently in additional storage location called secondary memory.

Output devices: The device that receives data from a computer system for display, physical production, etc. Is called output devices. It converts digital information into human understandable form. For example, monitor, projector, headphones, speaker, printer, etc.

Evolution of computer

From the simple calculator to a modern day powerful data processor, computing devices have evolved in relatively short span of time. The evolution of computing devices is shown through a timeline. In the following figure:



1
500 BC
|
Abacus

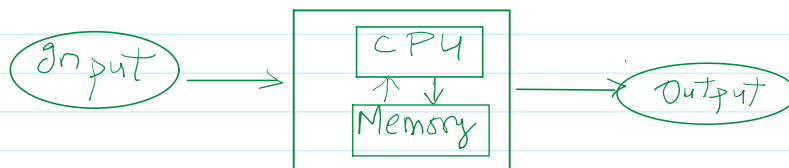
1834
|
Analytical Engine
(Charles Babbage)

1947
|
Transistors
[vacuum Tubes were
replaced by transistors
developed at Bell Labs,
using semiconductor
materials]

1937
|
Turing Machine

[concept was a general purpose
programmable machine that was capable of
solving any problem by executing the program
stored on the punched cards.]

Timeline showing key inventions in computing technology.



Von Neumann architecture for the computer

ENIAC [Electronic Numerical Integrated And Computer]

Computer Memory

Computer system needs memory to store. The data and instructions for processing. Whenever we talk about memory of a computer system, we usually talk about the main or primary memory. The secondary memory (also called storage device) is used to store data, instructions and results permanently for future use.

Units of Memory

A computer system uses binary numbers to store and process data. The binary digits zero and one, which are the basic units of memory, are called bits. Further, these bits are grouped together to form words. A 4 - bit word is called a nibble. Example of nibbles are 1001, 1010, 0010 etc. A 2 nibble word that is eight - bit word is called a byte. For example. 01000110, 01111100 etc.

Like any other standard, unit bytes are grouped together to make bigger chunks of units of memory. The following table shows different measurement units for digital data stored in a storage device.

Measurement units for digital data

Unit	Description
KB(Kilobyte)	1 KB = 1024 Bytes
MB(Megabyte)	1 MB = 1024 KB

GB(Gigabyte)	1 GB = 1024 MB
TB(Terabyte)	1 TB = 1024 GB
PB(Petabyte)	1 PB = 1024 TB
EB(Exabyte)	1 EB = 1024 PB
ZB(Zetabyte)	1 ZB = 1024 PB
YB(Yottabyte)	1 YB = 1024 ZB

Type of Memory

Are human beings memorize many things over a lifetime and recall from memory to make a decision or some action. However, we do not rely on our memory completely and we make notes and store important data and information using other media such as notebooks, manual, journal document, etc. Similarly, computers have two types of memory - primary and secondary.

1. Primary Memory: Primary memory is an essential component of a computer system. Program and data are loaded into the primary memory before processing. The CPU interacts directly with the primary memory to perform read and write operation. It is of two types.

1. Random Access Memory (RAM)

2. Read Only Memory. (ROM)

RAM: RAM is volatile. That is, as long as the power is supplied to the computer, it retains the data in it. But as soon as the power supply is turned off, all the contents of RAM are wiped out. It is used to store data temporarily while the computer is working. Whenever the computer is started or a software application is launched, the required program and data are loaded into RAM for processing. RAM is usually referred to as main memory and it is faster than the secondary memory or storage devices.

ROM: ROM is non-volatile, which means its contents are not lost even when the power is turned off. It is used as a small but faster permanent storage for the contents, which are rarely changed. For example, the start-up program (boot loader) that loads the operating system into primary memory is stored in ROM.

2. Cache Memory: RAM is faster than secondary storage, but not as fast as computer processor. So because of RAM, a CPU may have to slow down. To speed up the operations of the CPU, a very high speed memory is placed between the CPU and the primary memory known as cache. It stores the copies of the data from frequently accessed. Primary memory locations, thus reducing the average time required to access data from the primary memory. When the CPU needs some data, it first examines the cache. In case the requirement is met, it is read from the cache, otherwise the primary memory is accessed.

3. Secondary Memory: Primary memory has limited storage capacity and is either volatile (RAM) or read only memory (ROM). Does a computer system needs auxiliary or secondary memory to permanently store the data or instructions for a future use. The secondary memory is non-volatile and has a larger storage capacity than primary memory. It is slower and cheaper than the main memory, but it cannot be accessed directly by the CPU. Contents of secondary storage needs to be first brought into the main memory of the CPU to access. Examples of secondary memory devices include hard disk drive (HDD) CD / DVD memory card. Solid state device (SSD) etc.



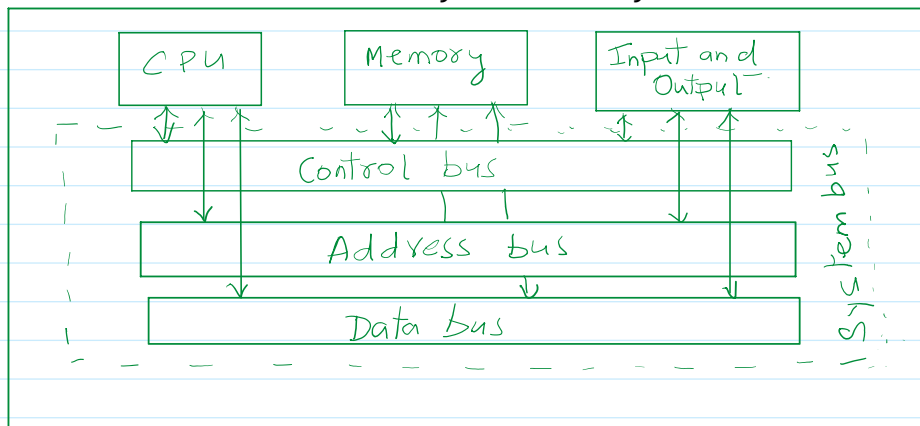
Data transfer between Memory and CPU

Data needs to be transferred between the CPU and primary memory, as well as between primary and secondary memory.

Data are transferred between different components of a computer system using physical wires called bus. For example, bus is used for data transfer between USB port and hard disk or between a hard disk and main memory. Bus is of three types.

- i) Data bus to transfer data between different components.
- ii) Address bus to transfer addresses between CPU and the main memory. The address of the memory location that the CPU wants to read or write from is specified in the address bus.
- iii) Control bus to communicate control signals between the different components of a computer.

All these three busses collectively make the system bus.



Data transfer between components through system bus.

As the CPU interacts directly with main memory, any data entered from the input device or the data to be accessed from hard disk or SSD needs to be placed in the main memory for further processor. The data is then transferred between CPU and main memory using bus.

The CPU places on the address bus the address of the main memory location from which it wants to read data or to write data. While executing the instructions, the CPU specifies the read or write control signal through the control bus.

As the CPU may require to read data from main memory or write data to main memory, a data bus is bidirectional. But the control bus and address bus are unidirectional. To write data into memory, the CPU places the data on the

data bus, which is then written to the specific address provided through the address bus. In case of read operation, the CPU specifies the address and the data is placed on the data bus by a dedicated hardware called memory controller. The memory controller manages the flow of data into and out of the computer's main memory.

Microprocessors

In early days, a computer CPU is used to occupy a large room or multiple cabinets. However with advance. In technology, the physical size of CPU has reduced and now possible to place a CPU on a single microchip only. A processor (CPU) which is implemented on a single microchip is called microprocessor. Nowadays, almost all Cpus are microprocessors. Hence the term are used synonymously for practical purposes.

Microprocessor is a small sized electronic component inside a computer that carries out various task involved in data processing as well as arithmetic and logical operations. These days a microprocessor is a built over an integrated circuit comprising millions of small components like registers, transistors and diodes.

Microprocessors have evolved overtime in terms of their increasing processing capability, decreasing physical size and reduced cost. Currently available microprocessors are capable of processing millions of instructions per second.

Here's a structured table showing the evolution of microprocessors across generations, including chip type, word size, maximum memory, clock speed, cores, and examples. This captures the major milestones from the 1970s to modern multi-core processors.

Microprocessor Generations Overview

Generation Era	Chip Type	Word Size	Max Memory Size	Clock Speed	Core s	Examples
First Gen (1971-1972)	SSI/MSI ICs	4-bit	~640 bytes	740 kHz	1	Intel 4004, Rockwell PPS-4
Second Gen (1973-1978)	NMOS	8-bit	64 KB	500 kHz – 3 MHz	1	Intel 8080, 8085, Motorola 6800
Third Gen (1979-1980)	HMOS	16-bit	1 MB	4-10 MHz	1	Intel 8086, 80286, Motorola 68000
Fourth Gen (1981-1995)	HMOS/CMOS	32-bit	4 GB	16-100 MHz	1	Intel 80386, 80486, Motorola 68020
Fifth Gen (1995-2005)	CMOS	64-bit	16 EB (theoretical)	66 MHz – 3 GHz	1-2	Intel Pentium, AMD Athlon
Modern Era (2006-Present)	CMOS, Multi-core	64-bit	>16 EB	1.2 GHz – 6 GHz	2-64 +	Intel Core i3/i5/i7/i9, AMD Ryzen, Apple M1/M2

Key Highlights

- **First Generation (Intel 4004, 1971):** 4-bit processor, only 640 bytes of memory, single-core at 740 kHz.
- **Second Generation (Intel 8085, 1976):** 8-bit, 64 KB memory, 3 MHz clock speed.
- **Third Generation (Intel 8086, 1978):** 16-bit, 1 MB memory, 10 MHz clock speed.

- **Fourth Generation (Intel 80386, 1986):** 32-bit, 4 GB memory, up to 33 MHz.
- **Fifth Generation (Pentium, 1993):** 64-bit, multi-core beginnings, GHz-level speeds.
- **Modern Era (Core i-series, Ryzen, Apple M1):** Multi-core (up to 64+), speeds beyond 5 GHz, massive memory addressing.

Risks & Trade-offs

- **Heat & Power Consumption:** Higher clock speeds and more cores increase thermal output, requiring advanced cooling.
- **Backward Compatibility:** Older software may not run efficiently on modern architectures.
- **Cost vs. Performance:** Cutting-edge chips (Apple M2 Ultra, Intel Core Ultra) are expensive compared to mid-range CPUs.

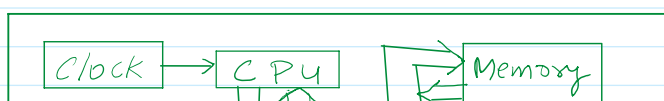
Microprocessor Specifications

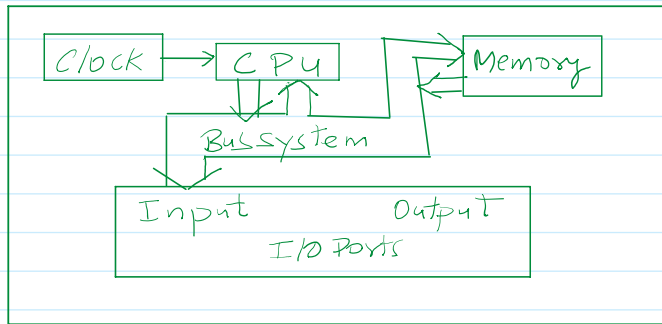
Microprocessors are classified on the basis of different features, include chip type, word size, memory size, clock speed, etcetera. These features are briefly explained below:

1. **Word size:** Word size is the maximum number of bits that a microprocessor can process at a time. Earlier a word was of eight bits as it was the maximum limit at that time. At present, the minimum word size is 16 bits and the maximum word size is 64 bits.
2. **Memory Size:** Depending upon the word size, the size of RAM varies. Initially RAM was very small (4MB,) Due to 4/8 bits word size. As word size increase to 64 bits, it has become feasible to use RAM of size up to 16 exabytes (EB)
3. **Clock speed:** Computers have internal clock that generates pulses (signals) at regular intervals of time. Clock speed simply means the number of pulses generated per second by the clock inside a computer. The clock speed indicates the speed at which the computer can execute instructions. Earlier it was measured in Hz(Hertz) and Kilohertz(kHz). But the advancement in technology and chip density? It is now measured in gigahertz (GHz), i.e., Billions of pulses per second.
4. **Cores:** Core is a basic computation unit of the CPU. Earlier processor had only one computation unit, thereby capable of performing only one task at a time. With the advent of multicore processors, it has become possible for the computer to execute multiple tasks, thereby increasing the systems performance. CPU with 2, 4 and eight cores are is called dual core, quad core and Octa core processors respectively.

Microcontrollers

The microcontroller is a small computing device which has a CPU of fixed amount of RAM, ROM and other peripherals all embedded on a single chip as compared to microprocessor that has only a CPU on the chip. The structure of a microcontroller is shown in the following figure:





Structure of microcontroller

Keyboard, mouse, washing machine, digital camera, pen drive, remote controller, microwave are few examples of microcontrollers. As these are designed for a specific task only, hence their sizes as well as cost is reduced.

Because of the very small size of the microcontroller, it is embedded in another device or system to perform a specific functionality. For example, the microcontroller in a fully automatic washing machine is used to control the washing cycle without any human intervention. The cycle starts with the filling of water, after which the clothes are soaked and washed. Thereafter the water is drained and the clothes are spin dry. The simple use of microcontroller has permitted repetitive execution of tedious task automatically without any human intervention, thereby saving precious time.

